LEARNER GUIDE

Surds: manipulating and rationalising

Number Systems and Calculator Judgement · Lesson 6. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A surd times itself is a whole number: $\sqrt{3} \times \sqrt{3} = 3$. With that, you can expand brackets with surds using a grid, collecting whole numbers and surds separately. In some grids the two surd cells are equal and opposite, so they cancel: $(3 + \sqrt{2})(3 - \sqrt{2}) = 9 - 2 = 7$. Brackets like these are called **conjugates**. They let you **rationalise the denominator**: multiply the numerator and the denominator by the right surd or conjugate, so the denominator becomes a whole number. Because you multiply by a fraction equal to 1, the value does not change.

What you are learning to notice

- $\sqrt{b} \times \sqrt{b} = b$. Multiply whole numbers together and surds together: $3\sqrt{2} \times 4\sqrt{2} = 12 \times 2 = 24$.
- Expand with a 2 by 2 grid, then collect: $(2 + \sqrt{3})(1 + \sqrt{3}) = 2 + 2\sqrt{3} + \sqrt{3} + 3 = 5 + 3\sqrt{3}$.
- Conjugates: $(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$. No surd is left.
- For $1/\sqrt{b}$, multiply by $\sqrt{b}/\sqrt{b}$. For $c/(a + \sqrt{b})$, multiply by the conjugate over itself.
- $(1 + \sqrt{2})^2$ is not $1 + 2$: squaring a bracket gives four cells.

Moving through it, one step at a time

To rationalise: look at the denominator. If it is $\sqrt{b}$, multiply the numerator and the denominator by $\sqrt{b}$. If it is $a + \sqrt{b}$ (or $a - \sqrt{b}$), multiply both by the conjugate. Work out the new denominator (it should be a whole number), multiply out the numerator, then simplify.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — The expansion grid

Expand $(3 + \sqrt{2})(3 - \sqrt{2})$.

1. Row 3: 3×3 and $3 \times (-\sqrt{2})$.	9 and $-3\sqrt{2}$
2. Row $\sqrt{2}$: $\sqrt{2} \times 3$ and $\sqrt{2} \times (-\sqrt{2})$.	$3\sqrt{2}$ and -2
3. The surd cells $-3\sqrt{2}$ and $+3\sqrt{2}$ are equal and opposite.	they cancel
4. Collect the whole numbers.	$9 - 2 = 7$

$(3 + \sqrt{2})(3 - \sqrt{2}) = 7$. No surd is left: these brackets are conjugates.

Example 2 — Rationalising with the conjugate

Rationalise the denominator of $4/(3 + \sqrt{5})$.

1. Multiply the numerator and the denominator by $3 – \sqrt{5}$.	this is multiplying by 1
2. Denominator: $(3 + \sqrt{5})(3 – \sqrt{5})$.	$= 9 – 5 = 4$
3. Numerator: $4(3 – \sqrt{5})$.	$= 12 – 4\sqrt{5}$
4. Divide each term by 4.	$3 – \sqrt{5}$

$4/(3 + \sqrt{5}) = 3 – \sqrt{5}$. Check: $(3 – \sqrt{5})(3 + \sqrt{5}) = 4$. ✓

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (expanding brackets, and simplifying surds), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.